

JOYITA FARUK

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RESEARCH INTERESTS

Computational Biology, AI/Deep Learning, Multi-modal Learning, Graph Neural Networks, Data Analytics

EDUCATION

Bachelor of Science | *Computer Science* | *CGPA 3.54/4.0*
Independent University, Bangladesh

Sep. 2019 – Jul 2024
Dhaka, Bangladesh

RESEARCH

Kidney Abnormality Detection Using Segmentation-Guided Classification on CT Images

Published in: International Conference on Machine Learning and Cybernetics (ICMLC), Miyazaki, Japan.
Pages 414–419. DOI: 10.1109/ICMLC63072.2024.10935113

Kidney Abnormal Region Localization on 3D CT Images Using ROI-Guided Segmentation

Undergraduate Thesis

WORK EXPERIENCE

Data Scientist, R&D Department
SimpliSolve

Jun 2025 – Present
Texas, US (Remote)

- Researching the latest methodologies to support the development of R&D projects
- Understanding and developing state-of-the-art AI models
- Conducting thorough exploratory data analysis, discerning patterns and insights
- Writing efficient and robust GPU-compatible Python code
- Deploying developed codes using cloud services such as AWS
- Leading an AI Innovators talent hunt at Independent University, Bangladesh on behalf of the company, designing the competition dataset, coordinating logistics by collaborating with marketing and academic teams.

Genomics Intern
Genereveal

Feb 2025 – Jun 2025
Montana, US (Remote)

- Worked on human DNA sequence data, exploring disease associations
- Leveraged Slurm-based HPC for large-scale genomic data processing
- Analyzed genomic data via alignment, variant calling, and tools like BWA, GATK, and SAMtools
- Developed and optimized bash scripts to automate workflows

SKILLS

Machine Learning & AI: Keras, TensorFlow, PyTorch, AI Agent Workflows

Cloud & DevOps: AWS, Docker, Terraform, Linux Shell Scripting

Programming Languages: Python, Java, C++

Database: MySQL, Firebase, Neo4j

Web & Software Development: Django, Flutter

Document Creation: $\LaTeX$, Markdown, Microsoft Office Suite

Languages: Bengali (Native), English (IELTS Academic 8.5), French (Preparing for A1)

HONORS AND AWARDS

Mitacs Globalink Research Award January 2025
Competitive award for conducting research internship in Canada; Grant Amount: \$6000 CAD (deferred due to program timing adjustments)

3-Minute Thesis Competition Winner March 2024
1st Place at the CSE Annual Research Day, Independent University, Bangladesh

Microprocessor Project Showcase Champion December 2023
1st Place at IUB Autumn Tech Fest, recognized for technical innovation and presentation

Web Application Project Showcase Winner July 2023
1st Place at IUB Summer Tech Fest for developing an innovative competition platform prototype

Vice Chancellor's List 2024
Awarded for attaining a CGPA of 3.50 or more in the third successive semester at Independent University, Bangladesh

Dean's Merit List 2024, 2022
Awarded for attaining a CGPA of 3.50 or more in the second successive semester at Independent University, Bangladesh

Dean's List 2023
Awarded for attaining a CGPA of 3.50 or more in any semester at Independent University, Bangladesh

Merit Scholarship 2019–2022
Awarded based on university entrance exam; received full or partial tuition coverage across multiple semesters

WORKSHOPS

Interdisciplinary Computational Biology Workshop 2025 January 2025
3-Day Workshop at BRAC University Dhaka, Bangladesh

2nd CCDS Workshop on Deep Learning Code Management December 2024
2-Day Workshop by CCDS, Independent University, Bangladesh Dhaka, Bangladesh

COURSE PROJECTS

TakaTracker – Expense Tracking Mobile Application 2024
Developed using Flutter and Firebase with features like currency conversion and categorized expense tracking

Cloud HemoSphere – Cloud-Based Blood Banking Web App 2023
Hosted on AWS using EC2 and VPC to ensure data security for blood bank management